

Fermions in a box

We now turn to an important application of our Fermionic formalism to many-body physics: the free Fermi gas. We consider a collection of N non-interacting spin-1/2 fermions of mass m . Despite having no interactions at all, the different fermions in this system behave in a nontrivial way due to the Pauli exclusion principle. Our goal will be to derive some global properties of this “gas” within the context of second quantization.



Fock space for fermions. Recall that the fermionic Fock space is $\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}^{\otimes N}$ and that creation and annihilation operators that create and annihilate fermions satisfy

$$\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}, \quad \{a_\alpha, a_\beta\} = \{a_\alpha^\dagger, a_\beta^\dagger\} = 0. \quad (1)$$

The vacuum $|0\rangle$ satisfies $a_\alpha |0\rangle = 0$ for all α , and successive application of creation operators builds the Fock basis. The occupation number of mode α is $\hat{n}_\alpha = a_\alpha^\dagger a_\alpha$. For fermions, its only possible eigenvalues are 0 and 1 by Pauli exclusion. We consider a finite size box, in which momenta are quantized (discrete spectrum).

Modes in a finite box. To make the problem concrete, we confine the system to a cube of side L and volume $V = L^3$ with periodic boundary conditions. Imposing periodic boundary conditions in each direction for plane-wave solutions $e^{i\mathbf{p}\cdot\mathbf{x}}$ forces the allowed momenta to lie on a lattice in momentum space:

$$\mathbf{p} = \frac{2\pi}{L} \mathbf{n}, \quad \mathbf{n} = (n_x, n_y, n_z), \quad (2)$$

with n_x, n_y, n_z integers. Consecutive lattice points are separated by $2\pi/L$ in each direction, so each mode occupies a volume

$$\Delta^3 p = \left(\frac{2\pi}{L}\right)^3 = \frac{(2\pi)^3}{V} \quad (3)$$

in momentum space. This small geometric fact is the key to everything that follows: as $V \rightarrow \infty$ the lattice becomes arbitrarily fine, and the sum over allowed momenta becomes an integral.

We will have to keep track of both momentum states as well as spin $\sigma \in \{\uparrow, \downarrow\}$. Let us introduce a fermionic annihilation operator $a_{\mathbf{p},\sigma}$ and its adjoint $a_{\mathbf{p},\sigma}^\dagger$:

$$\{a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}^\dagger\} = \delta_{\mathbf{p}\mathbf{p}'} \delta_{\sigma\sigma'}, \quad \{a_{\mathbf{p},\sigma}, a_{\mathbf{p}',\sigma'}\} = 0, \quad (4)$$

where $\delta_{\mathbf{p}\mathbf{p}'}$ is a Kronecker delta on the lattice (2), and the vacuum $|0\rangle$ satisfies $a_{\mathbf{p},\sigma} |0\rangle = 0$ for all $(\mathbf{p}, \sigma)$.

The Pauli exclusion principle is encoded in Eq. (4). From the anticommutator $\{a_{\mathbf{p},\sigma}^\dagger, a_{\mathbf{p},\sigma}^\dagger\} = 0$, we get $2(a_{\mathbf{p},\sigma}^\dagger)^2 = 0$, so $(a_{\mathbf{p},\sigma}^\dagger)^2 = 0$. No mode can be occupied more than once. We also define the occupation number $\hat{n}_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}^\dagger a_{\mathbf{p},\sigma}$ and note that

$$\hat{n}_{\mathbf{p},\sigma}^2 = a_{\mathbf{p},\sigma}^\dagger (a_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma}^\dagger) a_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}^\dagger (1 - \hat{n}_{\mathbf{p},\sigma}) a_{\mathbf{p},\sigma} = \hat{n}_{\mathbf{p},\sigma}, \quad (5)$$

where we used $a_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma}^\dagger = 1 - \hat{n}_{\mathbf{p},\sigma}$ and the fact that $\hat{n}_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}^\dagger \hat{n}_{\mathbf{p},\sigma} = 0$. An operator satisfying $\hat{n}^2 = \hat{n}$ is a projector, and its eigenvalues are 0 and 1. The two subspaces are the one where $(\mathbf{p}, \sigma)$ is occupied and its complement is the subspace where it is empty. This is all we will need to understand the Fermi gas.

Field operator and Hamiltonian. Following the general prescription of the previous lecture, the fermionic field operator in the box is

$$\hat{\psi}_\sigma(\mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} a_{\mathbf{p},\sigma}, \quad \hat{\psi}_\sigma^\dagger(\mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{p}} e^{-i\mathbf{p}\cdot\mathbf{x}} a_{\mathbf{p},\sigma}^\dagger. \quad (6)$$

Using Eq. (4) and the completeness relation $V^{-1} \sum_{\mathbf{p}} e^{i\mathbf{p} \cdot (\mathbf{x} - \mathbf{y})} = \delta_L^3(\mathbf{x} - \mathbf{y})$, where δ_L^3 is the periodic delta function in the lattice and becomes δ^3 in the infinite-volume limit. One can check that

$$\{\hat{\psi}_\sigma(\mathbf{x}), \hat{\psi}_{\sigma'}^\dagger(\mathbf{y})\} = \delta_{\sigma\sigma'} \delta_L^3(\mathbf{x} - \mathbf{y}), \quad \{\hat{\psi}_\sigma(\mathbf{x}), \hat{\psi}_{\sigma'}(\mathbf{y})\} = 0. \quad (7)$$

The single-particle kinetic energy $\hat{T}$ has matrix elements $\langle \mathbf{p} | \hat{T} | \mathbf{p}' \rangle = \omega_{\mathbf{p}} \delta_{\mathbf{p}\mathbf{p}'}$ with $\omega_{\mathbf{p}} = \mathbf{p}^2/(2m)$, so by our usual prescription the free Hamiltonian is diagonal in momentum

$$\hat{H} = \sum_{\mathbf{p}, \sigma} \omega_{\mathbf{p}} a_{\mathbf{p}, \sigma}^\dagger a_{\mathbf{p}, \sigma} = \sum_{\mathbf{p}, \sigma} \omega_{\mathbf{p}} \hat{n}_{\mathbf{p}, \sigma}. \quad (8)$$

The total particle-number operator is just the sum of each mode's occupation number:

$$\hat{N} = \sum_{\mathbf{p}, \sigma} a_{\mathbf{p}, \sigma}^\dagger a_{\mathbf{p}, \sigma} = \sum_{\mathbf{p}, \sigma} \hat{n}_{\mathbf{p}, \sigma}. \quad (9)$$

Both $\hat{H}$ and $\hat{N}$ are diagonal in the occupation-number basis: on any Fock state $|\{n_{\mathbf{p}, \sigma}\}\rangle$ with $n_{\mathbf{p}, \sigma} \in \{0, 1\}$,

$$\hat{H} |\{n_{\mathbf{p}, \sigma}\}\rangle = \left(\sum_{\mathbf{p}, \sigma} \hat{n}_{\mathbf{p}, \sigma} \omega_{\mathbf{p}} \right) |\{n_{\mathbf{p}, \sigma}\}\rangle, \quad \hat{N} |\{n_{\mathbf{p}, \sigma}\}\rangle = \left(\sum_{\mathbf{p}, \sigma} \hat{n}_{\mathbf{p}, \sigma} \right) |\{n_{\mathbf{p}, \sigma}\}\rangle. \quad (10)$$

This is all very reasonable. Specifying which modes are occupied completely determines the energy and particle number, very analogous to our harmonic oscillator toy model.



Verify $\{\hat{\psi}_\sigma(\mathbf{x}), \hat{\psi}_{\sigma'}^\dagger(\mathbf{y})\} = \delta_{\sigma\sigma'} \delta_L^3(\mathbf{x} - \mathbf{y})$ by substituting Eq. (6) and using Eq. (4). Interpret the fact that $\{\hat{\psi}_\sigma(\mathbf{x}), \hat{\psi}_{\sigma'}(\mathbf{y})\} = 0$ physically.

The ground state. We want the lowest-energy eigenstate of $\hat{H}$ subject to having exactly N particles. From Eq. (10), the total energy is $E = \sum_{\mathbf{p}, \sigma} n_{\mathbf{p}, \sigma} \omega_{\mathbf{p}}$, a sum over non-negative terms, each at most 1. To minimize it at fixed particle number, we simply fill the N modes with the smallest possible energy $\omega_{\mathbf{p}} = |\mathbf{p}|^2/(2m)$, which correspond to filling states with the smallest possible momentum $|\mathbf{p}|$.



This means we will fill our states up to some threshold $p_F = |\mathbf{p}_F|$, which we refer to as the **Fermi momentum**, defined by the requirement that exactly N modes are filled. The result is our ground state

$$|\Omega\rangle = \prod_{|\mathbf{p}| < p_F} \prod_{\sigma \in \{\uparrow, \downarrow\}} a_{\mathbf{p}, \sigma}^\dagger |0\rangle. \quad (11)$$

It is very important that the product of creation operators in Eq. (11) increases without skipping any momentum states. The exact ordering is less important since all the creation operators (all at different $\mathbf{p}$) anticommute and so the different orderings only differ by an unphysical sign.

Strictly speaking, for a finite box the last state may or may not be degenerate, but in the continuum (thermodynamic) limit this shell ambiguity is irrelevant. We then represent the filled states in the momentum space by the **Fermi sphere**.

Acting with $\hat{N}$ on the ground state should recover N . Each occupied mode $(\mathbf{p}, \sigma)$ contributes an eigenvalue 1 to the sum and we get an additional factor of 2 from the two spin states at each $\mathbf{p}$:

$$\hat{N} |\Omega\rangle = \left(\sum_{|\mathbf{p}| < p_F} \sum_{\sigma} \hat{n}_{\mathbf{p}, \sigma} \right) |\Omega\rangle = N |\Omega\rangle, \quad N = 2 \sum_{|\mathbf{p}| < p_F} 1. \quad (12)$$

Similarly, we can also easily calculate the ground state energy $\hat{H} |\Omega\rangle = E_0 |\Omega\rangle$ with

$$E_0 = 2 \sum_{|\mathbf{p}| < p_F} \omega_{\mathbf{p}}. \quad (13)$$

Going to the continuum limit for Eq. (12) and Eq. (13). For a smooth function $f(\vec{p})$ that varies slowly between lattice points, we have

$$\sum_{\vec{p}} f(\vec{p}) = \frac{V}{(2\pi)^3} \sum_{\vec{p}} f(\vec{p}) \Delta^3 p \xrightarrow{V \rightarrow \infty} \frac{V}{(2\pi)^3} \int d^3 p f(\vec{p}). \quad (14)$$

Note that this is proportional to V . When we take $V \rightarrow \infty$, we will therefore only be interested in density quantities like $n = N/V$ fixed (this is the thermodynamic limit), so Eq. (14) becomes well defined.

The Fermi momentum. Instead of explicitly stating the range of the sum as in Eq. (12), we take $f(\vec{p}) = \theta(p_F - |\vec{p}|)$ and apply Eq. (14) to the particle count in Eq. (12):

$$N = \frac{2V}{(2\pi)^3} \int \theta(p_F - |\vec{p}|) d^3 p = \frac{2V}{(2\pi)^3} 4\pi \int_0^{p_F} p^2 dp = \frac{V p_F^3}{3\pi^2}. \quad (15)$$

Rearranging,

$$\boxed{p_F = (3\pi^2 n)^{1/3}}, \quad (16)$$

where $n = N/V$ is the particle density. We can also interpret the number density a bit more:

$$n = p_F^3 / (3\pi^2) = \frac{4\pi(p_F/2\pi)^3}{3} \times 2 \quad (17)$$

gives the volume of the Fermi sphere in momentum space with radius p_F . As the density increases, the sphere grows to accommodate more states.

The Fermi energy is the kinetic energy of a particle sitting exactly at the Fermi surface:

$$\boxed{E_F \equiv \frac{p_F^2}{2m} = \frac{(3\pi^2 n)^{2/3}}{2m}}. \quad (18)$$

All modes with $\omega_{\mathbf{p}} < E_F$ are occupied in the ground state; all modes with $\omega_{\mathbf{p}} > E_F$ are empty. Recall that the chemical potential μ is the energy cost of adding one more particle to the system, and at $T = 0$ this is exactly the energy of the first unoccupied mode, which is $\mu = E_F$.

The electrons in the conduction band of copper have $n \simeq 8.5 \times 10^{28} \text{ m}^{-3}$. With that, we find

$$E_F = \frac{(3\pi^2 n)^{2/3}}{2m_e} \simeq 7.0 \text{ eV}.$$

The thermal energy at room temperature is $k_B T \simeq 0.026 \text{ eV}$. Since $E_F/k_B T \approx 270 \gg 1$, the electrons are degenerate to a very good approximation.



Everything above assumed $T = 0$. For finite temperature, the distribution function $f(\omega_{\mathbf{p}})$ is generalized to the **Fermi-Dirac distribution** (you can find derivations in any statistical mechanics textbook):

$$\langle \hat{n}_{\mathbf{p},\sigma} \rangle = f(\omega_{\mathbf{p}}) \equiv \frac{1}{e^{(\omega_{\mathbf{p}} - \mu)/k_B T} + 1}, \quad (19)$$

with $\mu = \mu(T)$ the chemical potential at temperature T (no longer equal to E_F). At finite T , the step is smeared over an energy window of width $\sim k_B T$ around $\mu \approx E_F$.

More generally, for fermions with spin s and degeneracy $g_s = 2s + 1$, the same derivation gives

$$N = \frac{g_s V p_F^3}{6\pi^2}, \quad p_F = \left(\frac{6\pi^2 n}{g_s} \right)^{1/3}. \quad (20)$$

For spin- $\frac{1}{2}$ ($g_s = 2$), Eq. (20) reduces to Eq. (16). For “spinless” fermions (for example, strongly polarized electrons), $g_s = 1$ and at the same density, p_F is larger by a factor of $2^{1/3}$. This is easy to understand: with only one spin configuration, the Fermi sphere must be larger to fit the same number of particles.

Ground-state energy. We apply the same replacement (14) to the ground-state energy (13):

$$E_0 = \frac{2V}{(2\pi)^3} \int \theta(p_F - |\mathbf{p}|) \frac{p^2}{2m} d^3\mathbf{p} = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi}{2m} \int_0^{p_F} p^4 dp = \frac{2V}{(2\pi)^3} \cdot \frac{2\pi p_F^5}{5m} = \frac{V p_F^5}{10\pi^2 m}. \quad (21)$$

Now we eliminate $V p_F^3$ using the number of particles $N = nV$, where $V p_F^3 = 3\pi^2 N$, finding

$$E_0 = \frac{p_F^2}{2m} \cdot \frac{1}{5} \cdot \frac{V p_F^3}{\pi^2 m} \cdot m = \frac{3}{5} \cdot N \cdot \frac{p_F^2}{2m} = \boxed{\frac{3}{5} N E_F}. \quad (22)$$

This necessarily had to be smaller than $N E_F$ since E_F is the energy of the first unoccupied mode. The fact that it is smaller than it by a factor of $3/5$ is related to the well-known result that the average of p^2 over a sphere of radius p_F is $\langle p^2 \rangle_{\text{sphere}} = \frac{3}{5} p_F^2$. We can also derive the energy density E_0/V as a function of n and m and show that it scales as $\varepsilon \propto n^{5/3}$ or $\varepsilon \propto n E_F$.

The degeneracy pressure. Note that because of the Pauli exclusion principle, the particles must have kinetic energy even at zero temperature, and so the Fermi gas can still exert pressure. As the volume decreases with fixed N , the number density increases and the Fermi momentum p_F must increase too by Eq. (16). To see this, we express E_0 as a function of N and V only. Substituting $p_F = (3\pi^2 N/V)^{1/3}$ into Eq. (21):

$$E_0 = \frac{V}{5\pi^2 m} \left(\frac{3\pi^2 N}{V} \right)^{5/3} = \frac{(3\pi^2)^{2/3}}{5m} \cdot \frac{N^{5/3}}{V^{2/3}} \cdot 3\pi^2 \cdot \frac{1}{3\pi^2} = \frac{3(3\pi^2)^{2/3}}{10m} \frac{N^{5/3}}{V^{2/3}}. \quad (23)$$

The zero-temperature pressure is $P = -\partial E_0 / \partial V|_N$. Differentiating Eq. (23):

$$P = -\left. \frac{\partial E_0}{\partial V} \right|_N = \frac{2}{3} \cdot \frac{E_0}{V} = \boxed{\frac{2}{5} n E_F \propto n^{5/3}}. \quad (24)$$

This is the degeneracy pressure. It has nothing to do with thermal agitation or interactions: it is a direct consequence of the Pauli exclusion principle forcing the particles into high-momentum modes. Note that the relation $P = \frac{2}{3}(E_0/V)$ in Eq. (24) is the same relation that holds for a classical ideal gas at any temperature, even though the physical origin is completely different here.



Polytropic equation of state and compact stars. A power-law relation between pressure and density is called a polytropic equation of state, $P \propto n^\gamma$. For the non-relativistic ideal Fermi gas, Eq. (24) gives $\gamma = 5/3$. The degeneracy pressure of electrons supports white dwarf stars against gravitational collapse. Since $P \propto n^{5/3} \propto R^{-5}$ while the gravitational pressure scales as R^{-4} , there is always a radius at which the two balance. At very high densities the electrons become relativistic, however, for which the same analysis gives $P \propto n^{4/3}$ (a softer equation of state with $\gamma = 4/3$). Beyond the Chandrasekhar mass $M_{\text{Ch}} \simeq 1.44 M_\odot$ there is no equilibrium and the star collapses, leading to a neutron star or black hole. The neutron star itself is then supported by the degeneracy pressure of neutrons rather than electrons.

Density of states. It is useful to organize the single-particle spectrum by energy rather than momentum. The **density of states** $g(\epsilon)$ is defined so that $g(\epsilon)d\epsilon$ is the number of single-particle states (counting both spin projections) with energy in the interval $[\epsilon, \epsilon + d\epsilon]$. Equivalently, $g(\epsilon) = d\mathcal{N}/d\epsilon$, where $\mathcal{N}(\epsilon)$ is the total number of states with energy at most ϵ .

In the continuum limit, $\mathcal{N}(\epsilon)$ counts modes inside the sphere of radius $p(\epsilon) = \sqrt{2m\epsilon}$:

$$\mathcal{N}(\epsilon) = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi}{3} p(\epsilon)^3 = \frac{V(2m\epsilon)^{3/2}}{3\pi^2}. \quad (25)$$

Differentiating with respect to ϵ and using $dp/d\epsilon = m/p(\epsilon)$:

$$g(\epsilon) = \frac{d\mathcal{N}}{d\epsilon} = \frac{V(2m)^{3/2}}{2\pi^2} \epsilon^{1/2} = \frac{Vm p(\epsilon)}{\pi^2}. \quad (26)$$

The density of states grows as $\sqrt{\epsilon}$, a hallmark of the three-dimensional free-particle spectrum. It reflects the fact that the number of momentum-space lattice points in a thin shell of radius p and thickness dp scales as $4\pi p^2 dp$, while the dispersion $\epsilon = p^2/(2m)$ converts dp to $d\epsilon$ with a factor $dp/d\epsilon = m/p$.

The value of $g(\epsilon)$ at the Fermi surface is particularly important. Using Eq. (26) with $\epsilon = E_F$ and then eliminating Vp_F using $Vp_F^3 = 3\pi^2 N$:

$$g(E_F) = \frac{Vm p_F}{\pi^2} = \frac{Vm p_F^3}{3\pi^2} \cdot \frac{3}{p_F^2} = \frac{3N}{2E_F}. \quad (27)$$

This quantity enters many important physical and thermodynamic properties of the Fermi gas. You will likely learn more about it in statistical mechanics or condensed matter courses.

Interactions. Turning on interactions, such as the two-body ones we saw earlier, changes the energetics of our system because the ground state must also minimize the interaction energy. We are no longer guaranteed that the minimum kinetic energy configuration (the Fermi sphere) minimizes the total energy. In addition, interactions can scatter particles into new momentum modes, and there is no obvious reason why the notion of a Fermi surface should still be useful. It turns out, however, that in many cases there is still a Fermi surface, but not for the original fermions themselves, but for quasiparticles that are collective modes with the same quantum numbers (charge e , spin $\frac{1}{2}$, momentum $\vec{p}$) as the original electrons, but with a “renormalized” effective mass m^* and a finite lifetime. This type of interacting system is called a **Fermi liquid**.

Generalizing to a d -dimensional box

The same analysis can be carried out in any spatial dimension d . It is quite straightforward to check that all formulae generalize to d dimensions by replacing our S^3 sphere in momentum space with an S^d sphere of volume:

$$V_d(p_F) = \frac{\pi^{d/2} p_F^d}{\Gamma(d/2 + 1)}. \quad (28)$$

Noting that $N = g_s V / (2\pi)^d \cdot V_d(p_F)$, we get

$$n = \frac{N}{V} = \frac{g_s}{(2\pi)^d} \cdot \frac{\pi^{d/2} p_F^d}{\Gamma(d/2 + 1)}, \quad P = \frac{2}{d} \cdot \frac{E_0}{V} = \frac{2}{d+2} n E_F \propto n^{1+2/d}. \quad (29)$$

Note how $p_F \propto n^{1/d}$ and $E_F \propto n^{2/d}$, so the Fermi energy grows more slowly with density as the spatial dimension increases. The polytropic index is $\gamma = 1 + 2/d$, so the equation of state becomes softer as the spatial dimension increases.

The density of states is also interesting to look at

$$g(\epsilon) = \frac{g_s V (2m)^{d/2} \epsilon^{d/2-1}}{2^d \pi^{d/2} \Gamma(d/2)}, \quad (30)$$

One special case is $d = 2$, where the density of states is constant,

$$g(\epsilon) = \frac{g_s m V}{2\pi} \quad \text{in 2D.} \quad (31)$$

This is a consequence of the fact that the number of modes in a thin shell of radius p and thickness dp scales as $2\pi p dp$, while the dispersion $\epsilon = p^2/(2m)$ converts dp to $d\epsilon$ with a factor $dp/d\epsilon = m/p$, giving a constant density of states.

Particle-hole excitations.

Before closing, we make a few brief comments to piggy back on the sea of Fermions we have just constructed. When studying the dynamics on top of this sea, it is convenient to redefine the “vacuum” to be $|\Omega\rangle$ itself, rather than the empty state $|0\rangle$. One then introduces particle operators that create particles above the Fermi surface and **hole** operators that remove particles from below the Fermi surface:

$$c_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}, \quad |\mathbf{p}| > p_F \quad d_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}^\dagger, \quad |\mathbf{p}| < p_F. \quad (32)$$

By construction, $c_{\mathbf{p},\sigma}$ tries to destroy a particle in a mode that is empty in $|\Omega\rangle$, and $d_{\mathbf{p},\sigma}$ tries to create a particle in a mode that is already full, so both annihilate the Fermi sea:

$$c_{\mathbf{p},\sigma} |\Omega\rangle = 0 \quad (|\mathbf{p}| > p_F), \quad d_{\mathbf{p},\sigma} |\Omega\rangle = 0 \quad (|\mathbf{p}| < p_F). \quad (33)$$

In other words, $|\Omega\rangle$ acts as a vacuum for particle-hole excitations. The operator $c_{\mathbf{p},\sigma}^\dagger = a_{\mathbf{p},\sigma}^\dagger$ adds a particle above the Fermi surface, while $d_{\mathbf{p},\sigma}^\dagger = a_{\mathbf{p},\sigma}$ removes a particle from below it, creating a hole.

It is useful to rewrite the Hamiltonian in terms of these operators. Splitting the sum over modes at the Fermi surface and using $a_{\mathbf{p},\sigma}^\dagger a_{\mathbf{p},\sigma} = 1 - d_{\mathbf{p},\sigma}^\dagger d_{\mathbf{p},\sigma}$ for $|\mathbf{p}| < p_F$ (from the anticommutation relations, $d_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma}^\dagger$ gives $d_{\mathbf{p},\sigma}^\dagger d_{\mathbf{p},\sigma} = a_{\mathbf{p},\sigma} a_{\mathbf{p},\sigma}^\dagger = 1 - \hat{n}_{\mathbf{p},\sigma}$), we get

$$\hat{H} = E_0 + \sum_{\substack{|\mathbf{p}| > p_F \\ \sigma}} \omega_{\mathbf{p}} c_{\mathbf{p},\sigma}^\dagger c_{\mathbf{p},\sigma} - \sum_{\substack{|\mathbf{p}| < p_F \\ \sigma}} \omega_{\mathbf{p}} d_{\mathbf{p},\sigma}^\dagger d_{\mathbf{p},\sigma}. \quad (34)$$

Every term in Eq. (34) is normal-ordered with respect to $|\Omega\rangle$, so that the Fermi sea is not disturbed. The ground-state energy E_0 is just a constant offset (see Eq. (13)). Note we can define a particle and hole number operator by $\hat{N}_c = \sum_{|\mathbf{p}| > p_F, \sigma} c_{\mathbf{p},\sigma}^\dagger c_{\mathbf{p},\sigma}$ and $\hat{N}_d = \sum_{|\mathbf{p}| < p_F, \sigma} d_{\mathbf{p},\sigma}^\dagger d_{\mathbf{p},\sigma}$, but with respect to the Hamiltonian $\hat{H}$ itself the hole term comes with a minus sign.

The ground state $|\Omega\rangle$ has no particle or hole excitations, so it has energy E_0 . A state with a single hole at momentum $\mathbf{p}$ below the Fermi surface has energy $E_0 - \omega_{\mathbf{p}}$ under $\hat{H}$, because it is obtained by removing one occupied particle from the Fermi sea. A state with a single particle at momentum $\mathbf{p}'$ above the Fermi surface has energy $E_0 + \omega_{\mathbf{p}'}$. If we insist on keeping the total particle number fixed, the elementary excitation is a particle-hole pair: remove a particle from an occupied mode $\mathbf{p}$ and place it in an empty mode $\mathbf{p}'$, giving energy $E_0 + \omega_{\mathbf{p}'} - \omega_{\mathbf{p}}$. This is positive because $|\mathbf{p}'| > p_F \geq |\mathbf{p}|$. In relativistic theory of the Dirac sea, holes in the negative-energy sea are interpreted as antiparticles (positrons, for example).